

SageMaker for Continuous Deployment

Instructor

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SageMaker Endpoints

Once your model is ready, you can deploy it to SageMaker using SageMaker Endpoints.



An endpoint includes:

- **SageMaker Model**
 - **Model Container:** Supports custom containers for flexibility.
 - **Model Artifacts from S3:** Fetches model data from Amazon S3.
 - **IAM Role Permissions:** Manages access controls.
 - **Network Settings:** Configures network preferences.

SageMaker Endpoints

Once your model is ready, you can deploy it to SageMaker using SageMaker Endpoints.



An endpoint includes:

- **SageMaker Endpoint Configurations**
 - **Provisioned:** Standard deployment setup.
 - **Provisioned - Async Invocation:** Handles asynchronous requests.
 - **Serverless:** Offers on-demand, scalable resources.

SageMaker Endpoints

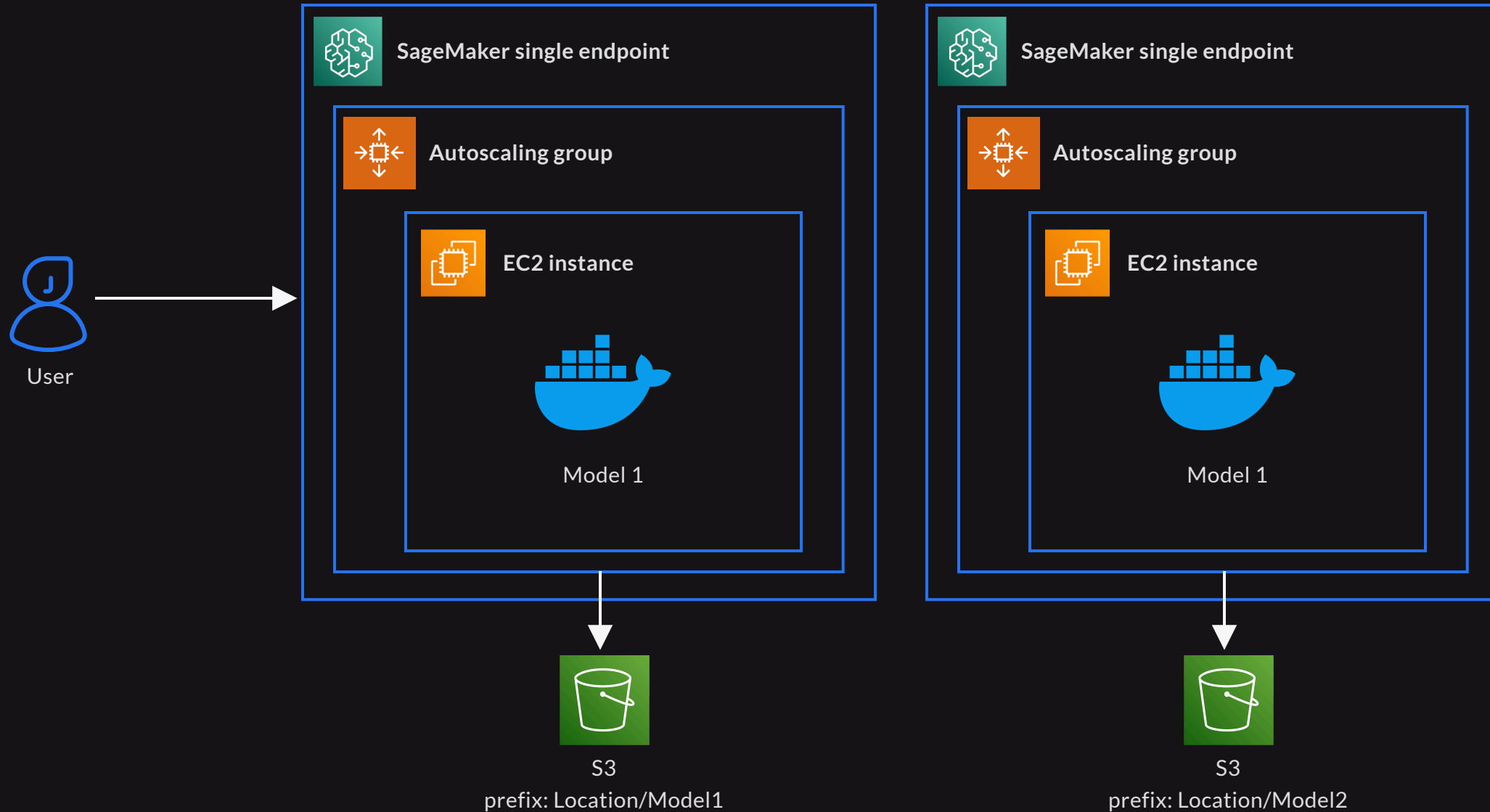
Once your model is ready, you can deploy it to SageMaker using SageMaker Endpoints.



An endpoint includes:

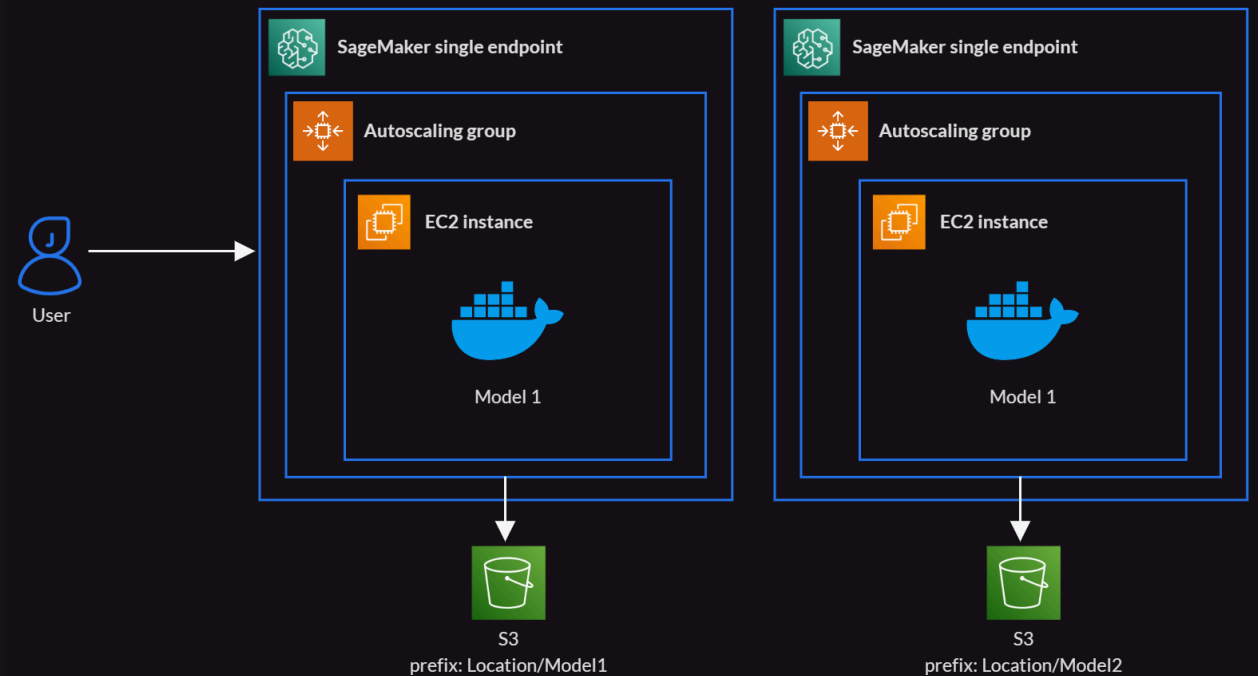
- **Additional Features**
 - **Model Card:** Share model details with your team and stakeholders.
 - **Data Capture:** Save input/output data to S3 for analysis and improvement.

Real-time Inference



Real-time Inference

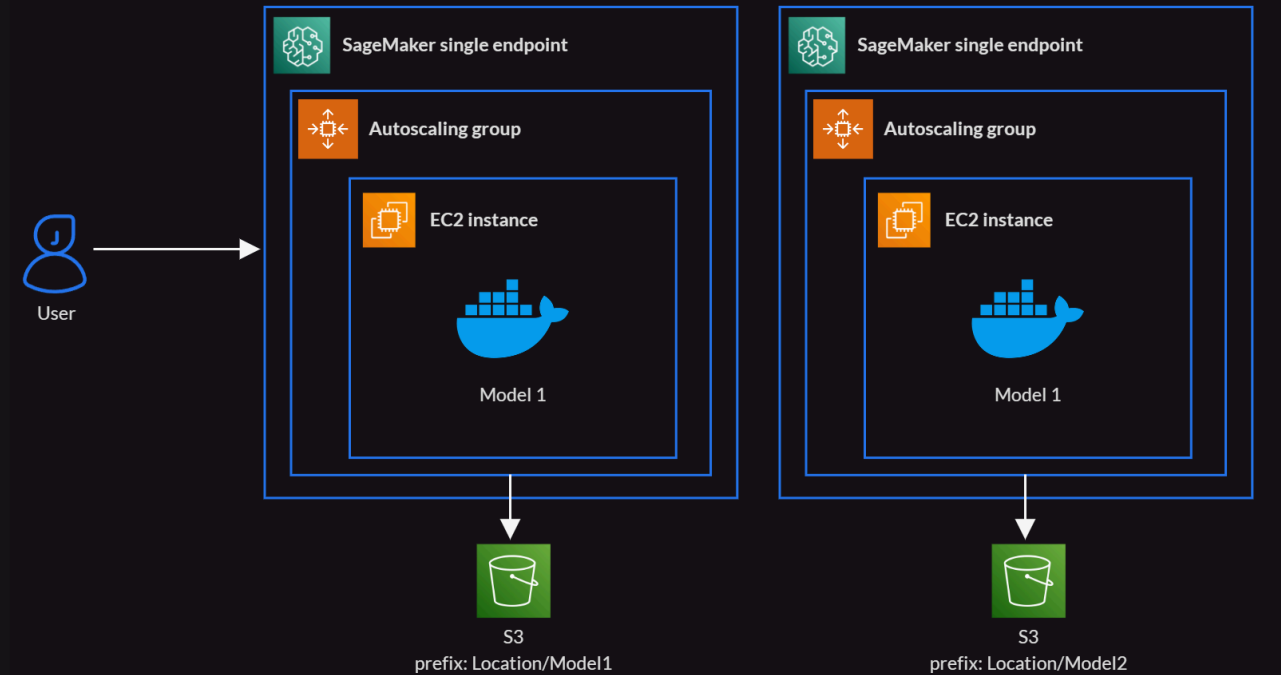
- Deploy models as web servers and keep them persistent
- In-build support for autoscaling
- Useful for high concurrency traffic in real-time
- Max 6 MB request payload and 60 MB response time
- Support for both CPU and GPU instances



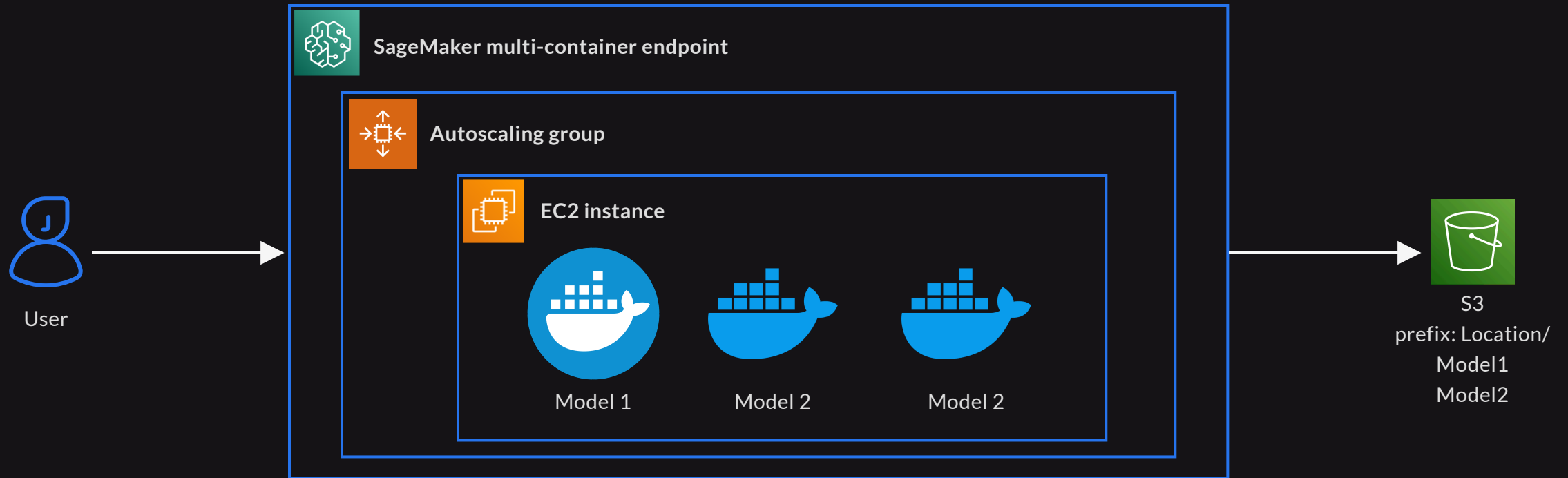
Real-time Inference

Hosting Types

- **Single model:** Deploy and scale a single model type on an instance



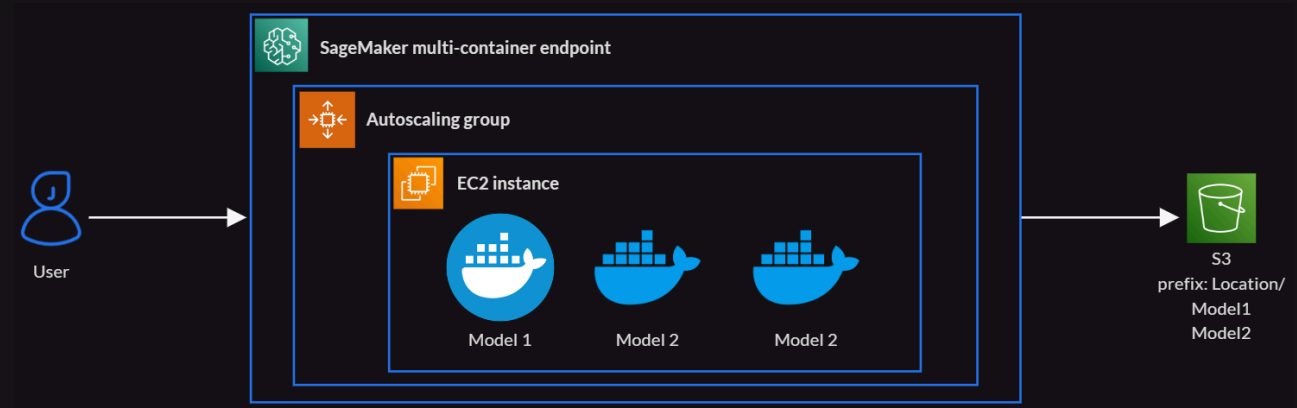
Real-time Inference



Real-time Inference

Multi-model in single endpoint

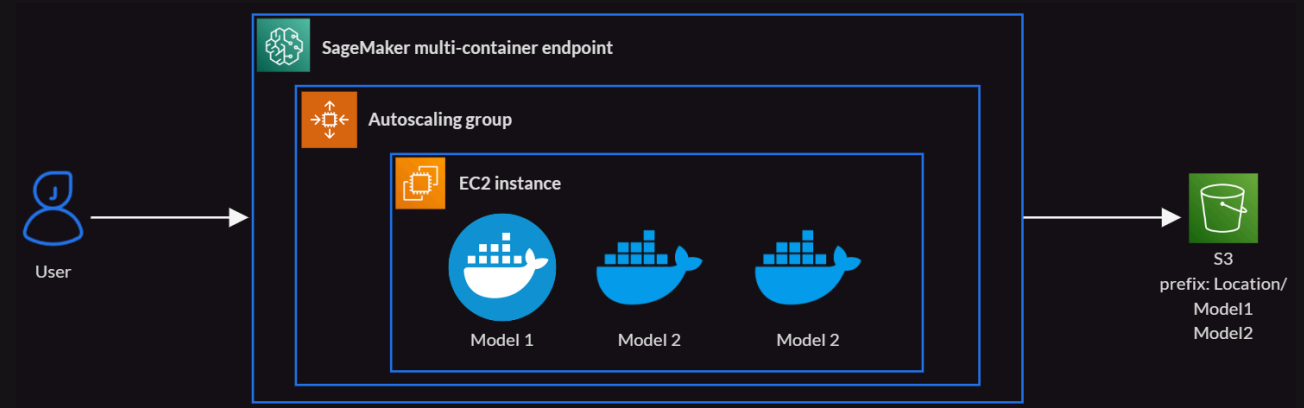
- Share instance resources between models
- Host similar models that share the exact same container image
- Different model artifacts are loaded from the same prefix folder
- The invocation specifies the target model to be invoked
- SageMaker auto-scales different model types based on invocation request patterns of these model
- Efficient GPU and CPU sharing saves cost



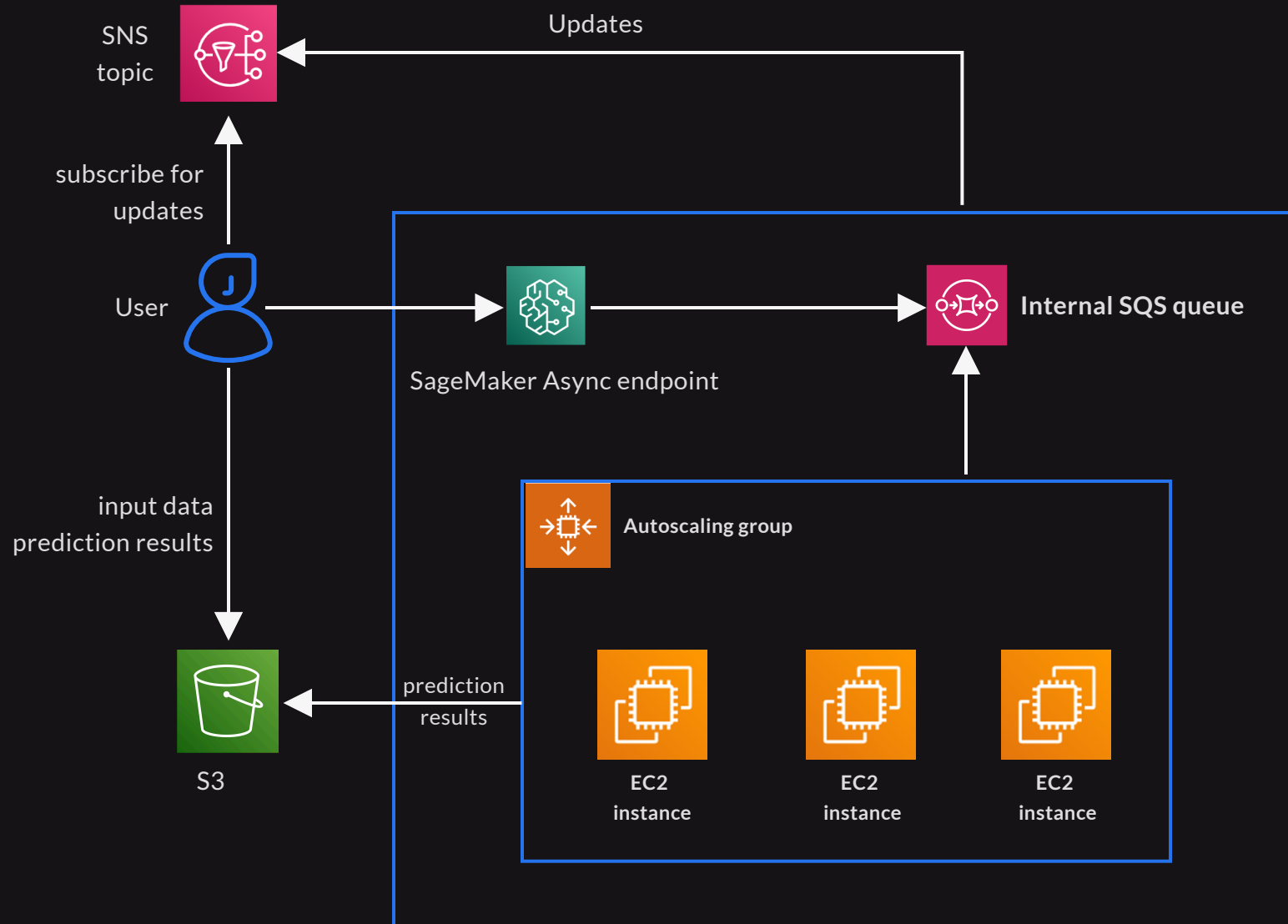
Real-time Inference

Multi-model with multi container in single endpoint

- Host upto 15 different types of containers on a single endpoint

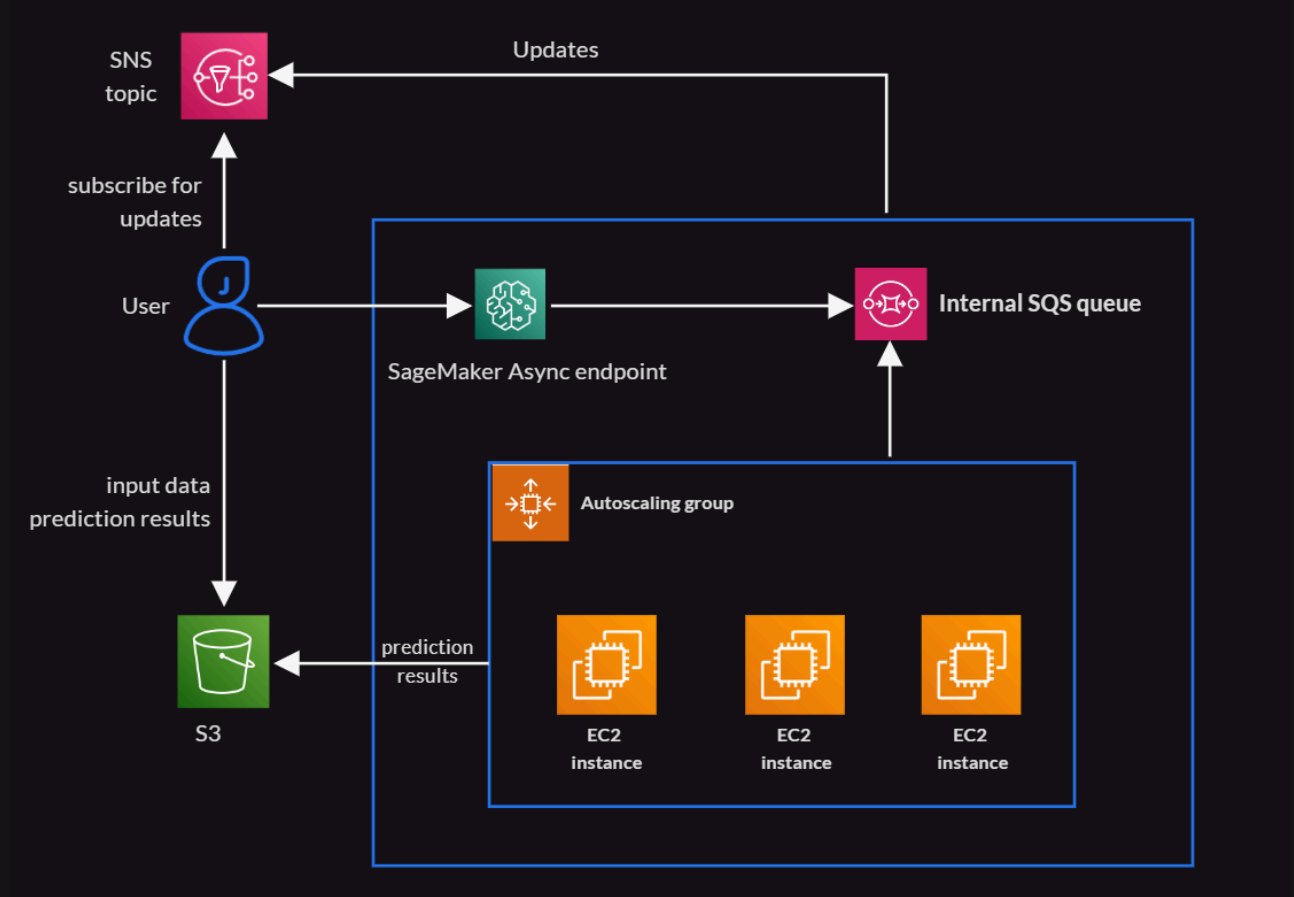


Asynchronous Inference

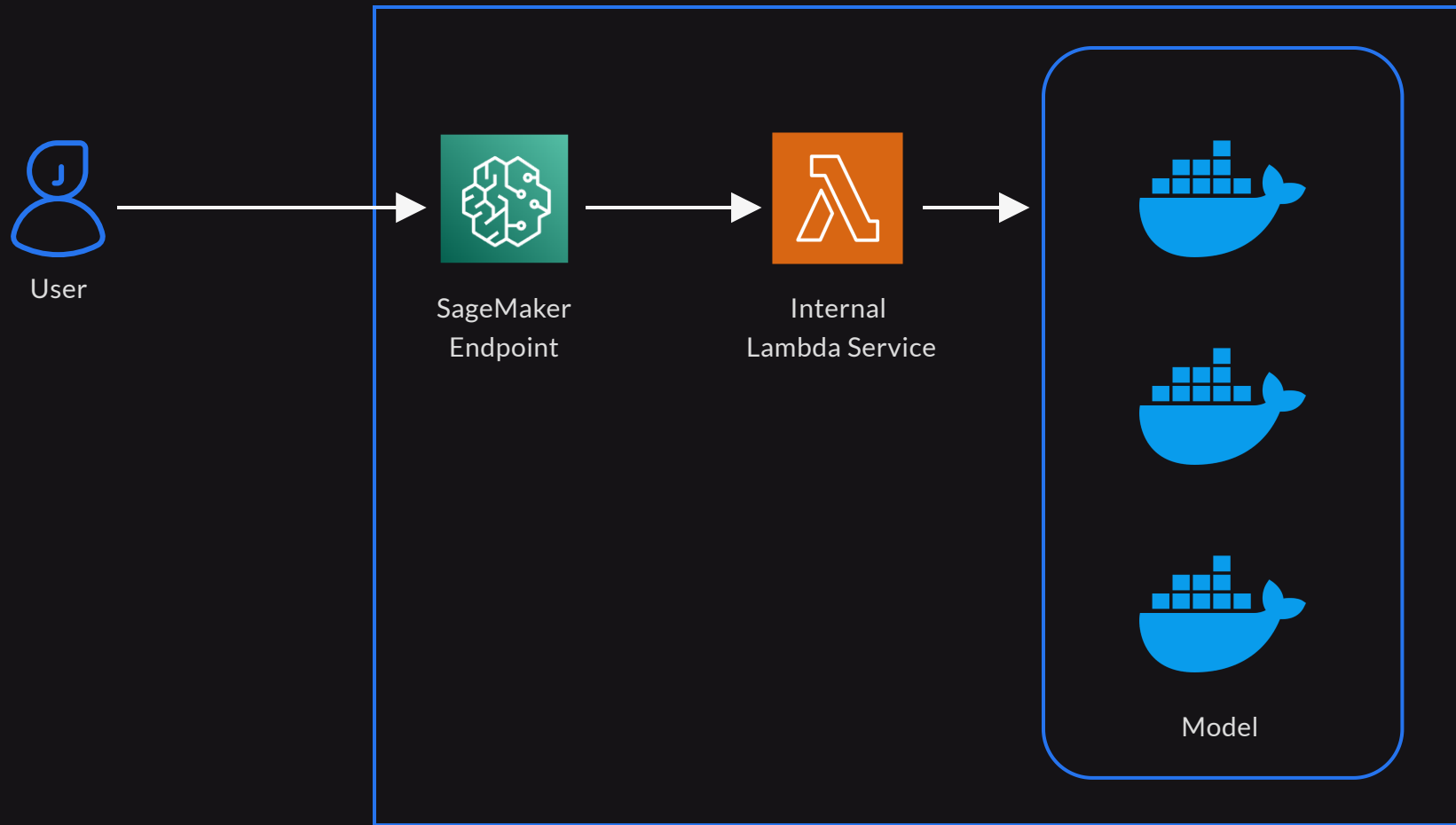


Asynchronous Inference

- Provision servers but call them asynchronously
- Requests with large payload sizes up to 1GB
- Long processing time up to 1 hour
- Near real-time latency
- Decoupled infrastructure using queues and notification topics
- Support for both CPU and GPU instances

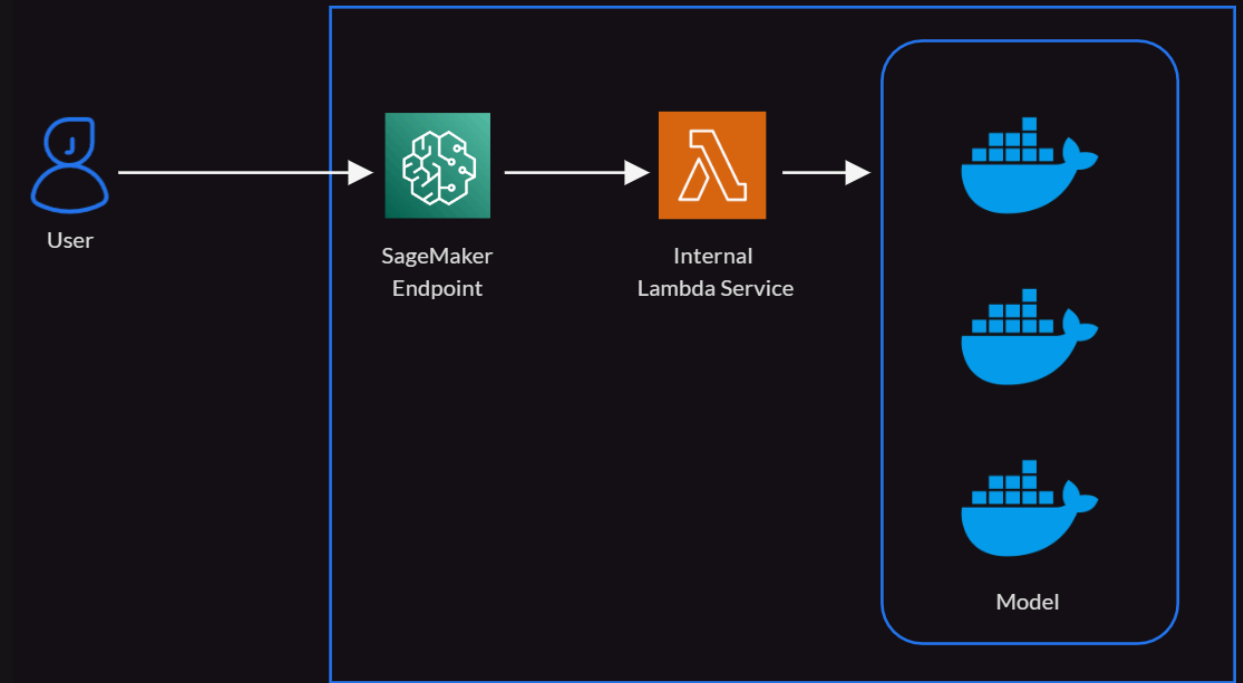


Serverless Inference



Serverless Inference

- Ideal for spiky traffic patterns with high concurrency limits
- Cold start time associated
- Fixed RAM values ranging from 1-6 GB maximum
- CPU scaled based on RAM
- No GPU support
- Container size of 10 GB with 5 GB additional storage
- Payload size of 4 MB with 60 seconds runtime
- Better to use custom AWS Lambda with up to 10 GB RAM with a 29-second runtime



Batch Transform Jobs

- To get predictions for an entire dataset
- No persistent endpoint. Servers launched based on set concurrency
- Distributed dataset feature also available
- Loads and saves output in S3
- Servers are destroyed upon completion



SageMaker Projects

Create end-to-end LLMOps / MLOps systems with 1 click deployment.

This ensures that we can:

- Fully automated using CloudFormation therefore not prone to human errors
- Enforce security best practices
- Setup templates based on organization needs
- Use AWS provided pre-built templates and modify them based on needs
- Share across teams in an organization

Cloud Formation Template

Code Repository

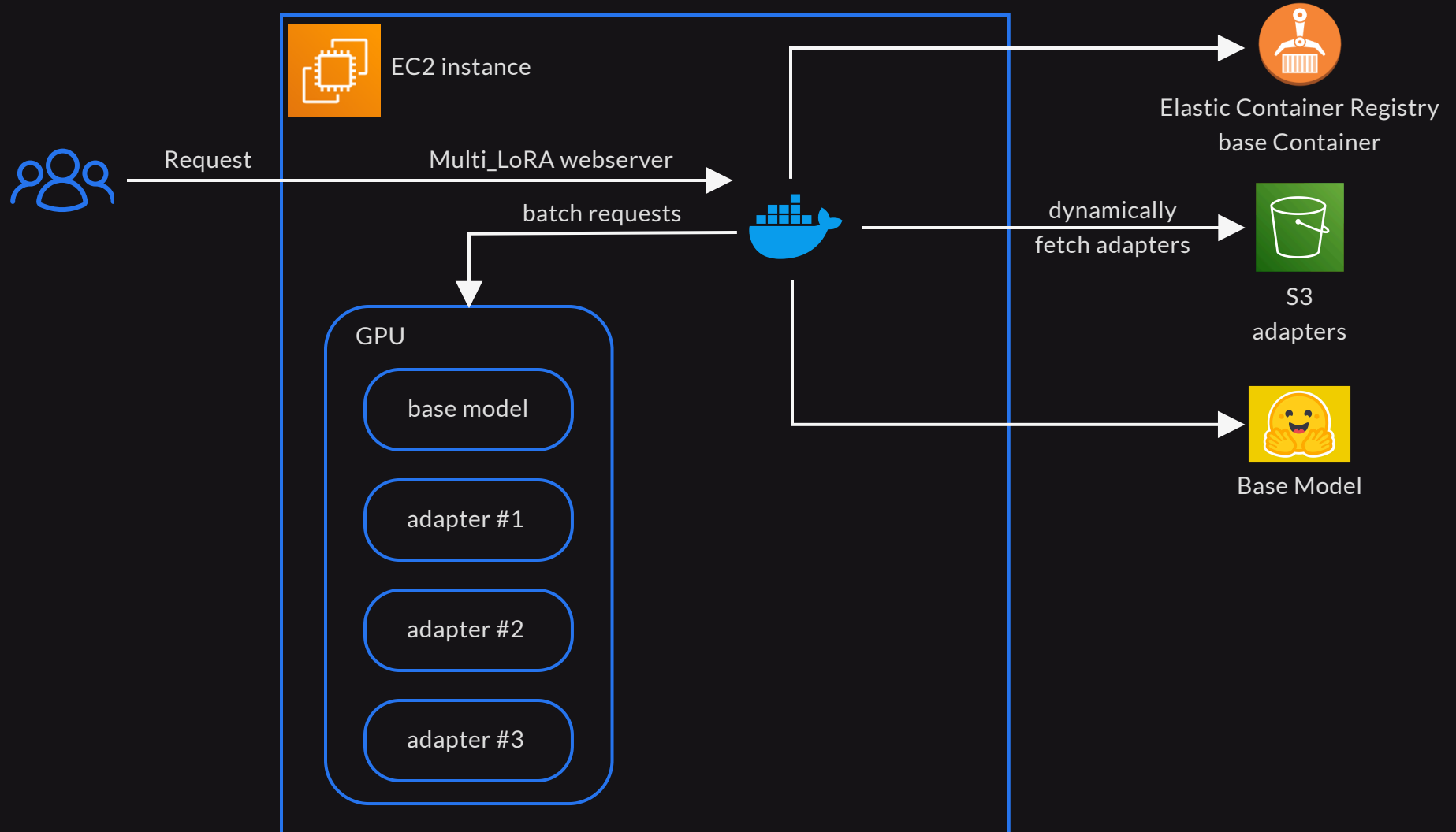
SageMaker Pipelines

SageMaker Experiments

Model Groups

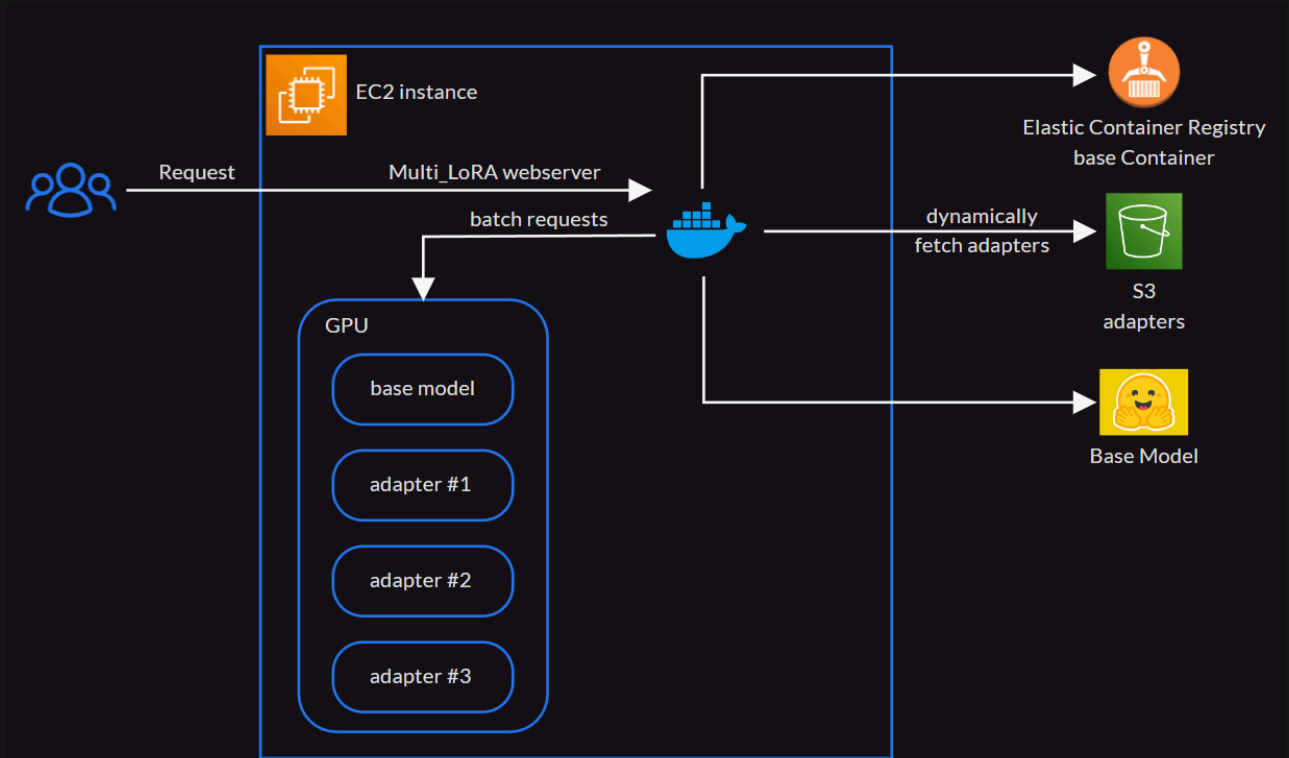
Endpoints

Multi-LoRA Serving



Multi-LoRA Serving

- Previously in SageMaker pipelines, we trained a LoRA adapter using a base model and SFT-based fine-tuning using the HuggingFace PEFT library
- We can have multiple adapters fine-tuned for different tasks
- Multi-LoRA serving dynamically loads and serves multiple LoRA adapters based on the incoming request
- The increase in memory for multiple adapters is small as every adapter is around 1-3% of model weights



Thank You
